

packages/graph-core/src/analyzer/normalize-graph.ts

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Overview

`packages/graph-core/src/analyzer/normalize-graph.ts` adds a set of utilities for normalizing graph data. The file introduces new functions, constants, and interfaces that operate on node IDs, complexity values, and batch-output structures.

Key changes

- **normalizeNodeId** – lines 64-110: normalizes IDs to `type:path`, handling double prefixes, project-name prefixes, and bare paths.
- **normalizeComplexity** – lines 134-152: maps string aliases and numeric scales to `"simple"` | `"moderate"` | `"complex"`.
- **normalizeBatchOutput** – lines 202-329: processes raw nodes/edges, fixes IDs, normalizes complexity, rewrites edge references, deduplicates nodes/edges, drops dangling edges, and returns a `NormalizeBatchResult`.
- **Interfaces** – `DroppedEdge` (lines 154-158), `NormalizationStats` (lines 161-166), `NormalizeBatchResult` (lines 169-173) provide structured output and metrics.
- **Helper constants** – `VALID_PREFIXES`, `TYPE_TO_PREFIX`, `VALID_COMPLEXITIES`, `COMPLEXITY_STRING_MAP`, `PREFIX_TO_TYPE` support the logic.
- **inferTypeFromId** – lines 184-192: derives a node type from an ID prefix, used during edge rewriting.

Impact

- **Correctness:** guarantees canonical IDs and complexity values before upstream pipelines, reducing downstream errors.
- **Observability:** `NormalizationStats` exposes counts of fixed IDs, complexity corrections, rewritten edges, and dropped edges.
- **Maintainability:** centralizes normalization logic; future changes to ID/complexity rules can be made in one place.
- **Performance:** operations use `Map` / `Set` lookups, yielding linear-time processing over nodes and edges.
- **Compatibility:** adds new exports but does not alter existing APIs; other modules can import these utilities without breaking changes.

Risks & follow-ups

- **Regression in ID handling:** verify that legacy double-prefixed IDs are still resolved correctly.
- **Edge deduplication:** ensure that deduplication does not drop legitimate parallel edges; run integration tests on graphs with intentional duplicates.

- **Stats accuracy:** cross-check `NormalizationStats` against manual counts on sample datasets.
- **Cross-module interactions:** confirm that downstream consumers of `normalizeBatchOutput` correctly handle the new `idMap` and `stats` fields.